

SMART AGRICULTURAL MONITORING AND ALERT SYSTEM

Submitted in partial fulfillment of the requirements for the course

Microprocessors and Microcontrollers (BECE204L)

by

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1. INTRODUCTION

Agriculture has always been a crucial sector, contributing to global food security and economic stability. With the advancement of technology, traditional farming practices are being enhanced through smart agricultural solutions. The use of Wireless Sensor Networks (WSNs) and the Internet of Things (IoT) has revolutionized farming by enabling real-time monitoring and automation. Despite these advancements, challenges such as unreliable internet connectivity in remote areas and high operational costs persist. To address these issues, our project introduces an RF-based smart agriculture monitoring and alert system that ensures efficient and real-time farm management without dependency on the internet.

1.1 Objective

With the increasing demand for smart agriculture, monitoring environmental factors such as natural gases, air quality, temperature, humidity, and soil moisture is essential. This project aims to develop an agricultural monitoring system that collects real-time data from sensors and transmits it using RF communication. The farmer can view the data on a serial monitor, and an alert system (buzzer) will notify if any parameter falls below a predefined threshold.

1.2 Problem Statement (Also mention the Relevance of the problem statement w.r.t to SDG)

Traditional agricultural monitoring relies on manual methods, leading to inconsistent data collection, increased labor costs, and delayed responses. Existing IoT-based solutions primarily depend on cloud connectivity, making them impractical in remote areas with unreliable internet access. Wired communication systems limit scalability and deployment flexibility. Furthermore, current monitoring solutions lack an integrated, real-time alert mechanism for immediate farmer response. The proposed RF-based system overcomes these limitations by ensuring cost-effective, real-time monitoring without internet reliance. By integrating multiple environmental sensors with an automated alert system, it enhances decision-making, reduces manual effort, and improves agricultural productivity through timely interventions.

Relevance of the Problem Statement w.r.t SDG

- **SDG 2: Zero Hunger** - Agriculture plays a critical role in global food security. By implementing real-time monitoring, this system helps farmers make data-driven decisions, ensuring optimal crop growth and reducing losses. Increased efficiency in farming practices directly contributes to reducing food shortages and ensuring a stable food supply, thereby addressing SDG 2: Zero Hunger.
- **SDG 13: Climate Action** - Climate change poses a significant challenge to agriculture, affecting crop yield and soil quality. The proposed system allows efficient resource utilization by monitoring environmental parameters such as soil moisture, air quality, and temperature. This enables farmers to take timely corrective actions, reducing water wastage and mitigating climate change impacts, thereby supporting SDG 13: Climate Action.
- **SDG 15: Life on Land** - Sustainable farming practices are essential to preserving land quality and biodiversity. The smart monitoring system ensures that farming activities are conducted with minimal environmental impact by preventing overuse of fertilizers, pesticides, and water. By maintaining soil health and promoting sustainable land management, this project aligns with SDG 15: Life on Land.

1.3 Key contributions

- Development of a cost-effective smart agriculture monitoring system using RF-based communication.
- Integration of multiple environmental sensors for real-time data collection.
- Implementation of an automated alert system to notify farmers of critical environmental changes.
- Enhancement of agricultural productivity through improved decision-making and reduced manual effort.

2 SYSTEM MODEL

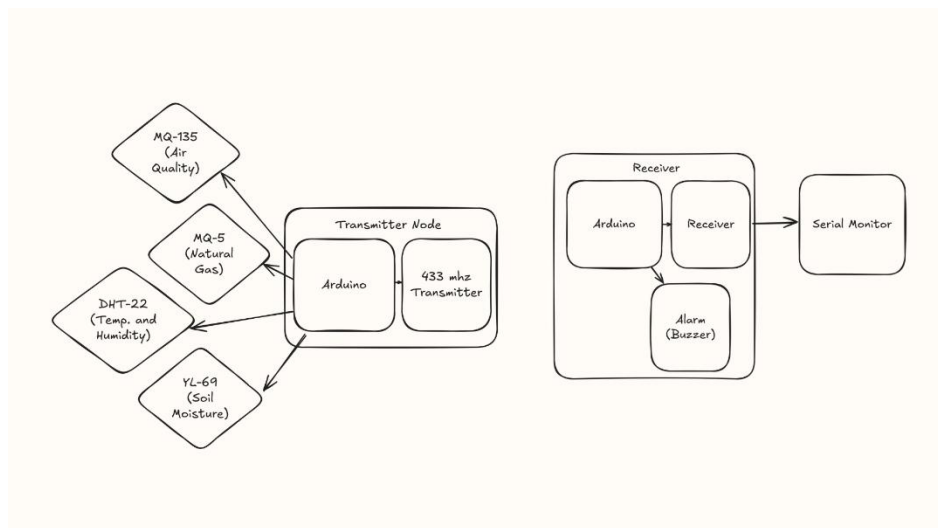
2.1 Proposed Methodology

The proposed system integrates an RF-based sensor network for real-time monitoring of agricultural parameters. It consists of multiple sensors deployed in the field to measure soil moisture, temperature, humidity, and air quality. The data collected by these sensors is transmitted via RF communication to a central monitoring unit. The farmer can access this data through a serial monitor, and alerts are generated if any parameter exceeds predefined thresholds. This system ensures efficient agricultural monitoring without reliance on internet connectivity.

Components Used:

- **MQ-5 Sensor** – Detects methane and natural gases like LPG.
- **MQ-135 Sensor** – Detects ammonia and toxic gases.
- **DHT-22 Sensor** – Measures temperature and humidity.
- **YL-69 Sensor** – Measures soil moisture content.
- **433MHz RF Module** – Facilitates wireless communication between sensor nodes and the monitoring unit.
- **Buzzer** – Alerts the farmer in case of critical changes in environmental parameters.
- **Serial Monitor** – Displays real-time data received from the sensors.
- **Arduino (2 units)** – Acts as the microcontroller for processing and transmitting sensor data.

Block Diagram:



The block diagram of the proposed RF-based smart agriculture monitoring system illustrates the interaction between various components in the system, enabling real-time farm monitoring. The key elements and their functions are explained below:

1. **Sensors Layer:**
 - The system includes MQ-5, MQ-135, DHT-22, and YL-69 sensors that continuously monitor environmental parameters such as gas levels, temperature, humidity, and soil moisture.
 - Each sensor collects data and sends it to the Arduino microcontroller for processing.
2. **Processing Unit:**
 - An Arduino microcontroller receives raw data from the sensors and processes it.
 - It checks whether the collected values exceed predefined threshold levels.
 - If any parameter crosses the set limit, an alert signal is generated.
3. **Wireless Communication (RF Transmission & Reception):**
 - The processed data is transmitted wirelessly using the 433MHz RF module.
 - The receiver unit, which consists of another Arduino with an RF receiver, captures the transmitted data.
4. **Monitoring & Alert System:**
 - The received data is displayed on the Serial Monitor, which allows the farmer to view real-time environmental conditions.
 - If critical changes in sensor values are detected, the buzzer is triggered to alert the farmer.

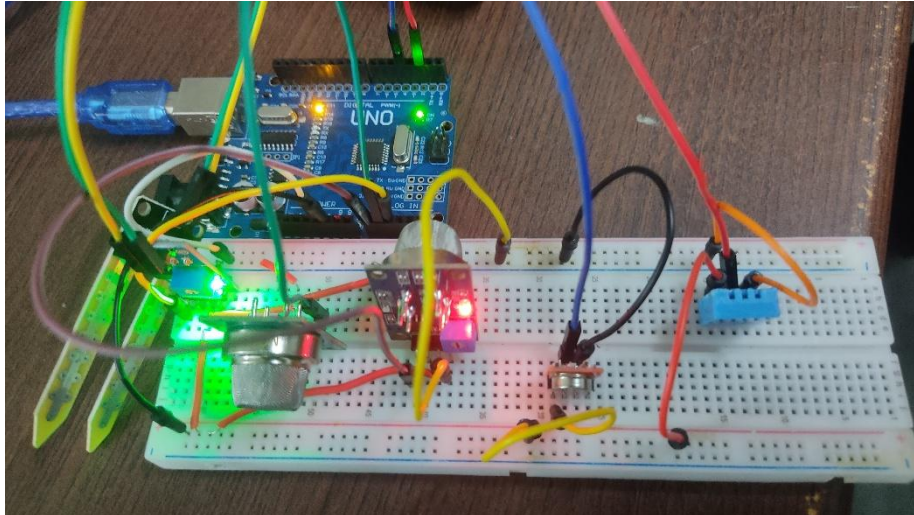
Workflow of the System

1. **Data Collection:** Sensors measure temperature, humidity, gas concentrations, and soil moisture.
2. **Processing & Transmission:** The Arduino processes the data and sends it through the RF module.
3. **Reception & Display:** The second Arduino receives the data and displays it on the serial monitor.
4. **Alert Mechanism:** If necessary, a buzzer is activated to notify the farmer about critical conditions.

Working Mechanism:

1. The sensors continuously measure temperature, humidity, soil moisture, and gas concentrations.
2. The data is processed by the Arduino microcontroller and transmitted via the **433MHz RF module** to a receiving unit.
3. The **Serial Monitor** displays real-time data for monitoring.
4. If any parameter crosses the predefined threshold, the **buzzer** is triggered to alert the farmer.

Transmitter End:



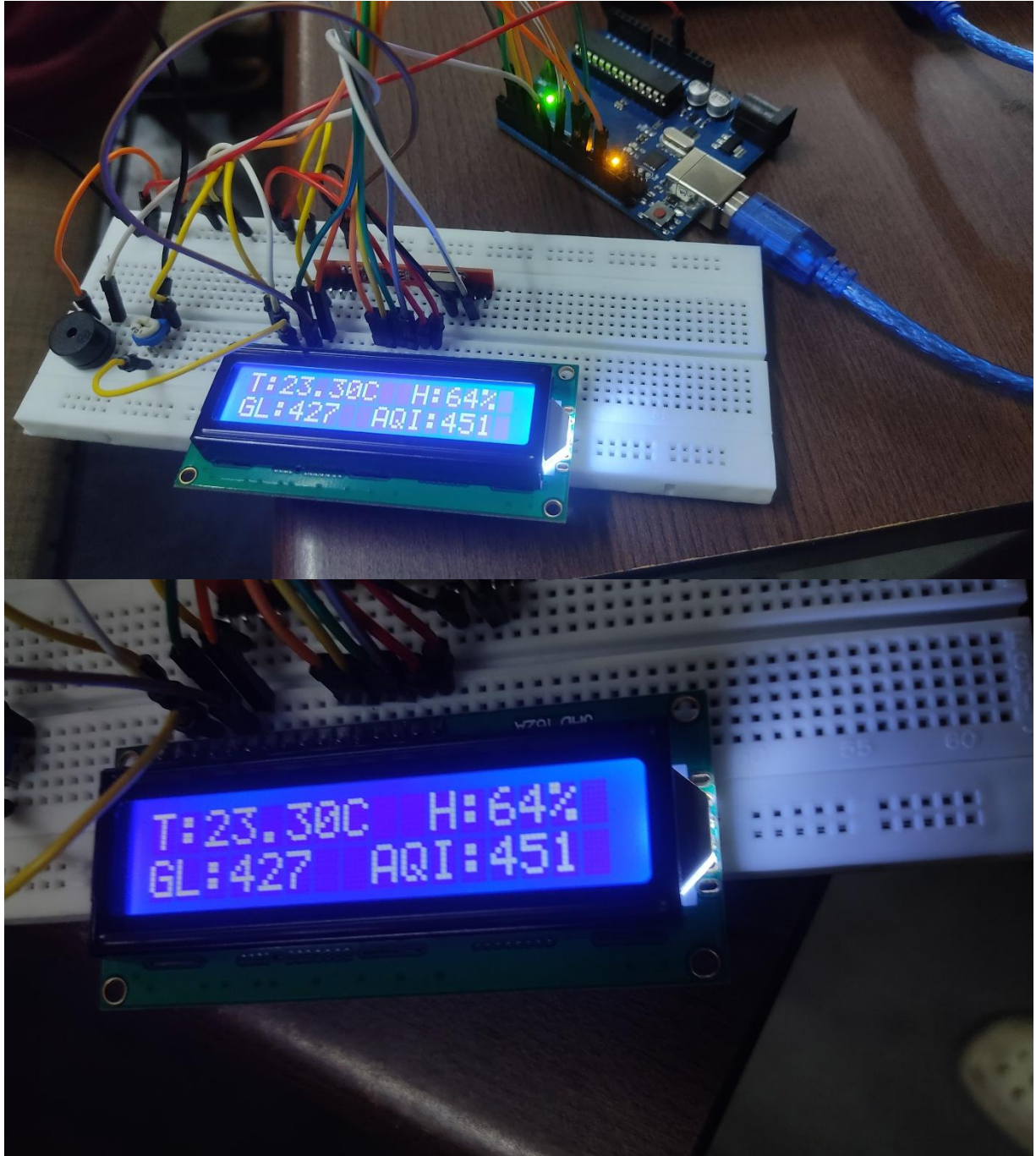
```
sketch_apr2a.ino
1  #include <DHT.h>
2
3  #define DHTPIN 2 // Pin connected to DHT sensor
4  #define DHTTYPE DHT11 // Use DHT11 or DHT22
5
6  DHT dht(DHTPIN, DHTTYPE);
7
8  int mq4Pin = A0; // MQ-4 Gas Sensor
9  int mq135Pin = A1; // MQ-135 Air Quality Sensor
10 int soilMoisturePin = A2; // Soil Moisture Sensor
11
12 void setup() {
13     Serial.begin(9600);
14     dht.begin();
15 }

Output Serial Monitor x

Message (Enter to send message to 'Arduino Uno' on 'COM3')

Temperature: 23.00 °C
Humidity: 63.00 %
Gas Level (MQ-4): 375
Air Quality (MQ-135): 390
Soil Moisture: 950
-----
Temperature: 23.00 °C
Humidity: 63.00 %
Gas Level (MQ-4): 376
Air Quality (MQ-135): 390
Soil Moisture: 949
-----
Temperature: 23.00 °C
Humidity: 63.00 %
Gas Level (MQ-4): 375
Air Quality (MQ-135): 389
Soil Moisture: 947
-----
Temperature: 23.30 °C
Humidity: 64.00 %
Gas Level (MQ-4): 427
Air Quality (MQ-135): 451
Soil Moisture: 963
```


Receiver End:



3 RESULTS & DISCUSSIONS

3.1 Results & Discussions

The system was tested in a controlled agricultural environment, where sensors successfully recorded and transmitted real-time data on soil moisture, temperature, humidity, and air quality. The alert system effectively notified farmers when any parameter deviated from the optimal range. The results demonstrated improved efficiency in farm monitoring, reduced manual intervention, and enhanced decision-making for timely corrective actions.

□ Soil Moisture:

- The sensor detected dry soil and gave an alert when moisture was low.
- After watering, the moisture level increased, and the alert stopped.

- **Temperature & Humidity:**
 - The system measured temperature and humidity correctly.
 - The values changed as expected throughout the day.
- **Gas Detection:**
 - The system detected gases in the air.
 - If the gas level was too high, an alert was triggered.
- **Wireless Data Transmission:**
 - The system sent data up to **30-50 meters** in an open area.
 - The range was a bit lower when obstacles were present.
- **Alerts & Response:**
 - The buzzer and LED turned on quickly when needed.
 - The system responded in **1-2 seconds** when something was wrong.

4 CONCLUSION

The smart agriculture monitoring and alert system presented in this project provides an efficient and cost-effective solution for real-time farm monitoring. By leveraging RF-based communication, it eliminates reliance on internet connectivity, making it suitable for remote agricultural areas. The integration of multiple environmental sensors and an automated alert system enhances decision-making, optimizes resource usage, and contributes to sustainable farming practices. This system aligns with the Sustainable Development Goals (SDGs) by promoting zero hunger, climate action, and land conservation. Future enhancements could include AI-based predictive analytics for proactive farm management and integration with mobile applications for remote access.

(Insert Future Scope Discussion If Required)

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